

Fig. 3.101

24*. In figure 3.102, two circles intersect each other at points A and E. Their common secant through E intersects the circles at points B and D. The tangents of the circles at points B and D intersect each other at point C. Prove that □ ABCD is cyclic.

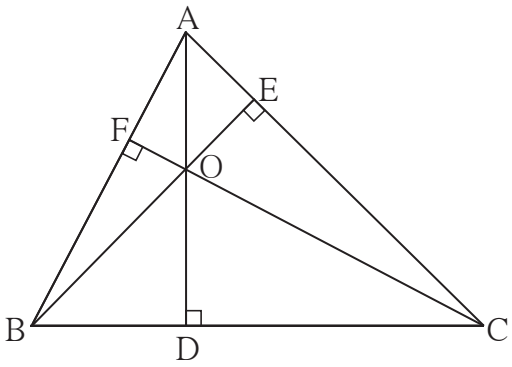


Fig. 3.103

23*. In figure 3.101, two circles intersect at points M and N. Secants drawn through M and N intersect the circles at points R, S and P, Q respectively.
Prove that : seg SQ || seg RP.

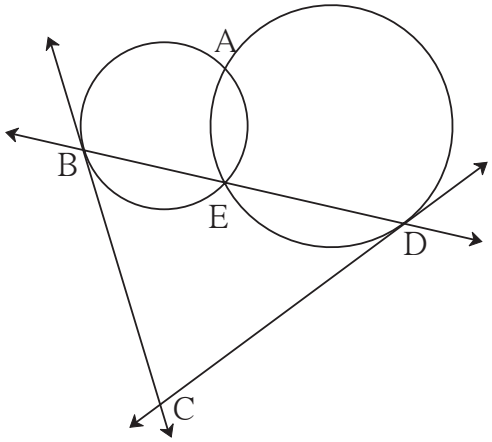


Fig. 3.102

25*. In figure 3.103, seg AD ⊥ side BC, seg BE ⊥ side AC, seg CF ⊥ side AB. Ponit O is the orthocentre. Prove that , point O is the incentre of Δ DEF.

ICT Tools or Links

Use the geogebra to verify the properties of chords and tangents of a circle.

PUEAJJ

4 Geometric Constructions



Let's study.

- Construction of a triangle similar to the given triangle
  - ✻ To construct a triangle, similar to the given triangle, bearing the given ratio with the sides of the given triangle.
    - (i) When vertices are distinct
    - (ii) When one vertex is common
- Construction of a tangent to a circle.
  - ✻ To construct a tangent at a point on the circle.
    - (i) Using centre of the circle.
    - (ii) Without using the centre of the circle.
  - ✻ To construct tangents to the given circle from a point outside the circle.



Let's recall.

In the previous standard you have learnt the following constructions. Let us recall those constructions.

- To construct a line parallel to a given line and passing through a given point outside the line.
- To construct the perpendicular bisector of a given line segment.
- To construct a triangle whose sides are given.
- To divide a given line segment into given number of equal parts
- To divide a line segment in the given ratio.
- To construct an angle congruent to the given angle.

In the ninth standard you have carried out the activity of preparing a map of surroundings of your school. Before constructing a building we make its plan. The surroundings of a school and its map, the building and its plan are similar to each other. We need to draw similar figures in Geography, architecture, machine drawing etc. A triangle is the simplest closed figure. We shall learn how to construct a triangle similar to the given triangle.

Let's learn.

Construction of Similar Triangle

To construct a triangle similar to the given triangle, satisfying the condition of given ratio of corresponding sides.

The corresponding sides of similar triangles are in the same proportion and the corresponding angles of these triangles are equal. Using this property, a triangle which is similar to the given triangle can be constructed.

Ex. (1) $\Delta ABC \sim \Delta PQR$, in ΔABC, AB = 5.4 cm, BC = 4.2 cm, AC = 6.0 cm. AB : PQ = 3 : 2. Construct ΔABC and ΔPQR.

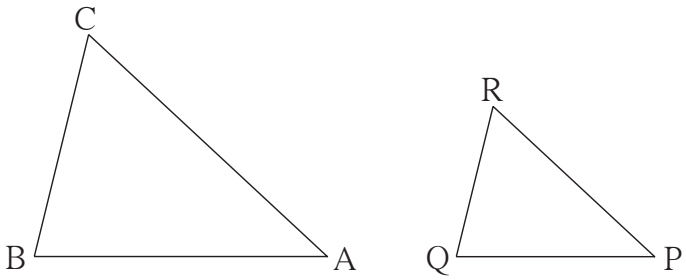


Fig. 4.1
Rough Figure

Construct ΔABC of given measure.

ΔABC and ΔPQR are similar.

$\therefore$ their corresponding sides are proportional.

$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR} = \frac{3}{2} \quad \ldots\ldots\ldots \text{(I)}$$

As the sides AB, BC, AC are known, we can find the lengths of sides PQ, QR, PR.

Using equation [I]

$$\frac{5.4}{PQ} = \frac{4.2}{QR} = \frac{6.0}{PR} = \frac{3}{2}$$

$\therefore$ PQ = 3.6 cm, QR = 2.8 cm and PR = 4.0 cm

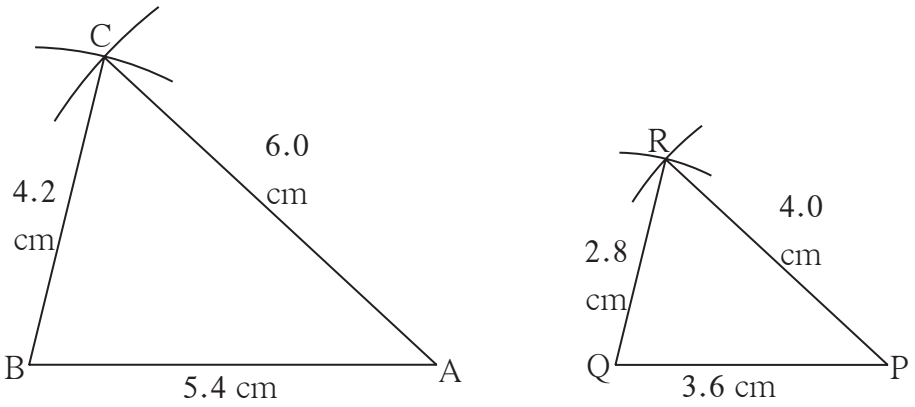


Fig. 4.2

For More Information

While drawing the triangle similar to the given triangle, sometimes the lengths of the sides we obtain by calculation are not easily measureable by a scale. In such a situation we can use the construction 'To divide the given segment in the given number of eqaul parts'.

For example, if length of side AB is $\frac{11.6}{3}$ cm, then by dividing the line segment of length 11.6 cm in three equal parts, we can draw segment AB.

If we know the lengths of all sides of Δ PQR, we can construct Δ PQR.

In the above example (1) there was no common vertex in the given triangle and the triangle to be constructed. If there is a common vertex, it is convenient to follow the method in the following example.

Ex.(2) Construct any Δ ABC. Construct Δ A′BC′ such that AB : A′B = 5:3 and Δ ABC ~ Δ A′BC′

Analysis : As shown in fig 4.3 , let the points B, A, A′ and B, C, C′ be collinear.

Δ ABC ~ Δ A′BC′ ∴ ∠ ABC = ∠A′BC′

$\frac{AB}{A'B} = \frac{BC}{BC'} = \frac{AC}{A'C'} = \frac{5}{3}$

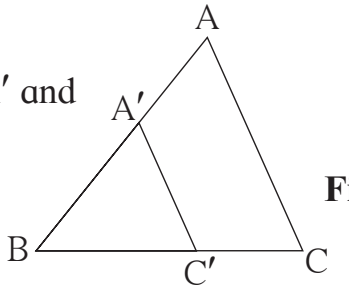


Fig. 4.3

Rough Figure

∴ sides of Δ ABC are longer than corresponding sides of Δ A′BC′ .

∴ the length of side BC′ will be equal to 3 parts out of 5 equal parts of side BC. So if we construct Δ ABC, point C′ will be on the side BC, at a distance equal to 3 parts from B. Now A′ is the point of intersection of AB and a line through C′, parallel to CA.

$$\frac{BA'}{BA} = \frac{BC'}{BC} = \frac{3}{5} \text{ i.e, } \frac{BA}{BA'} = \frac{BC}{BC'} = \frac{5}{3} \text{ ..... Taking inverse}$$

Steps of construction :

(1) Construct any Δ ABC.

(2) Divide segment BC in 5 equal parts.

(3) Name the end point of third part of seg BC as C′ ∴ $BC' = \frac{3}{5}$ BC

(4) Now draw a line parallel to AC through C′. Name the point where the parallel line intersects AB as A′.

(5) Δ A′BC′ is the required trinangle similar to Δ ABC

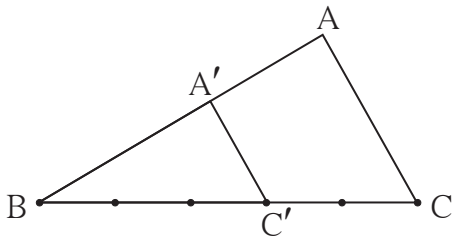


Fig. 4.4

Note : To divide segment BC, in five equal parts, it is convenient to draw a ray from B, on the side of line BC in which point A does not lie.

Take points T_1, T_2, T_3, T_4, T_5 on the ray such that

$BT_1 = T_1T_2 = T_2T_3 = T_3T_4 = T_4T_5$

Join T_5C and draw lines parallel to T_5C through T_1, T_2, T_3, T_4 .

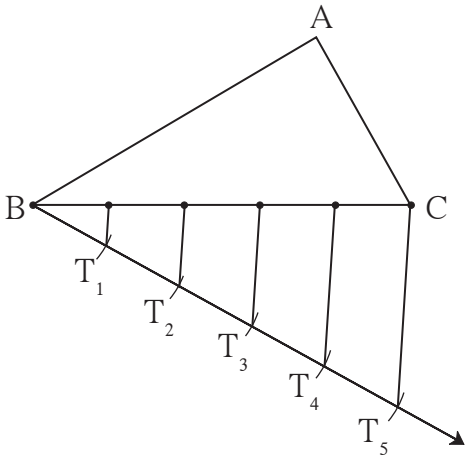


Fig. 4.5

Let's think

Δ A′BC′ can also be constructed as shown in the adjoining figure.

What changes do we have to make in steps of construction in that case ?

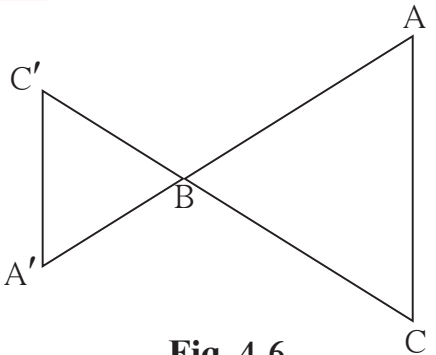


Fig. 4.6

Ex. (3) Construct $\Delta A'BC'$ similar to ΔABC such that $AB:A'B = 5:7$

Analysis : Let points B, A, A′ as well as points B, C, C′ be collinear.

$\Delta ABC \sim \Delta A'BC'$ and $AB : A'B = 5:7$

$\therefore$ sides of ΔABC are smaller than sides of $\Delta A'BC'$

and $\angle ABC \cong \angle A'BC'$

Let us draw a rough figure with these considerations. Now $\frac{BC}{BC'} = \frac{5}{7}$

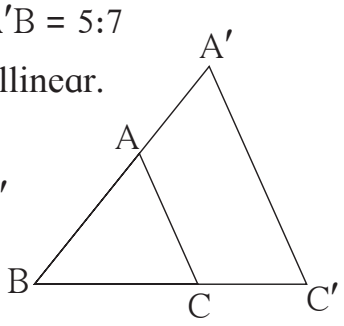


Fig. 4.7
Rough Figure

$\therefore$ If seg BC is divided into 5 equal parts, then seg BC′ will be 7 times each part of seg BC.

$\therefore$ let us divide side BC of ΔABC in 5 equal parts and locate point C′ at a distance equal to 7 such parts from B on ray BC. A line through point C′ parallel to seg AC is drawn it will intersect ray BA at point A′. According to the basic proportionality theorem we will get $\Delta A'BC'$ as described.

Steps of construction :

(1) Construct any ΔABC.

(2) Divide segment BC into 5 five equal parts. Fix point C′ on ray BC such that length of BC′ is seven times of each equal part of seg BC

(3) Draw a line parallel to side AC, through C′. Name the point of intersection of the line and ray BA as A′.

We get the required $\Delta A'BC'$ similar to ΔABC.

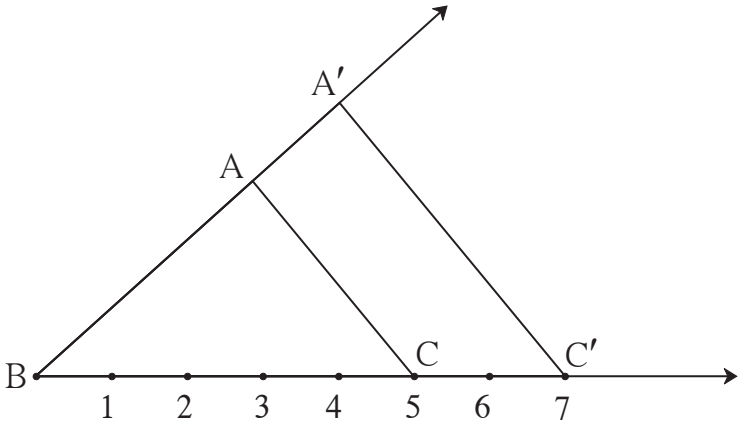


Fig. 4.8

Practice set 4.1

1. Δ ABC ~ Δ LMN. In Δ ABC, AB = 5.5 cm, BC = 6 cm, CA = 4.5 cm. Construct Δ ABC and Δ LMN such that $\frac{BC}{MN} = \frac{5}{4}$.

2. Δ PQR ~ Δ LTR. In Δ PQR, PQ = 4.2 cm, QR = 5.4 cm, PR = 4.8 cm. Construct Δ PQR and Δ LTR, such that $\frac{PQ}{LT} = \frac{3}{4}$.

3. Δ RST ~ ΔXYZ. In Δ RST, RS = 4.5 cm, $\angle$RST = 40°, ST = 5.7 cm Construct Δ RST and Δ XYZ, such that $\frac{RS}{XY} = \frac{3}{5}$.

4. Δ AMT ~ Δ AHE. In Δ AMT, AM = 6.3 cm, $\angle$TAM = 50°, AT = 5.6 cm. $\frac{AM}{AH} = \frac{7}{5}$. Construct Δ AHE.

Construction of a tangent to a circle at a point on the circle

(i) Using the centre of the circle.

Analysis :

Suppose we want to construct a tangent l passing through a point P on the circle with centre C. We shall use the property that a line perpendicular to the radius at its outer end is a tangent to the circle. If CP is a radius with point P on the circle, line l through P and perpendicular to CP is the tangent at P. For this we will use the construction of drawing a perpendicular to a line through a point on it.

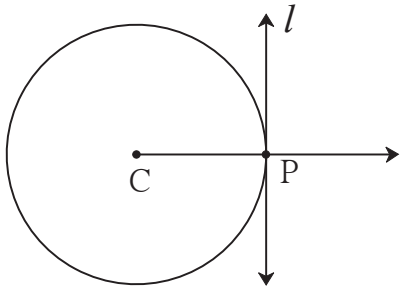


Fig. 4.9

For convenience we shall draw ray CP

Steps of construction

(1) Draw a circle with centre C.
Take any point P on the circle.

(2) Draw ray CP.

(3) Draw line l perpendicular to ray CX through point P.
Line l is the required tangent to the circle at point 'P'.

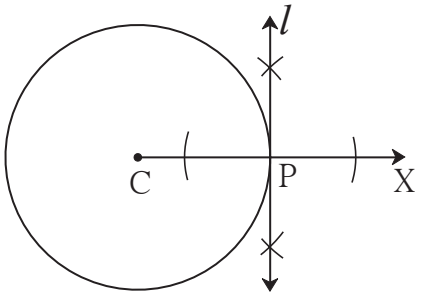


Fig. 4.10

ii) Without using the centre of the circle.

Example: Construct a circle of any radius. Take any point C on it. Construct a tangent to the circle without using centre of the circle.

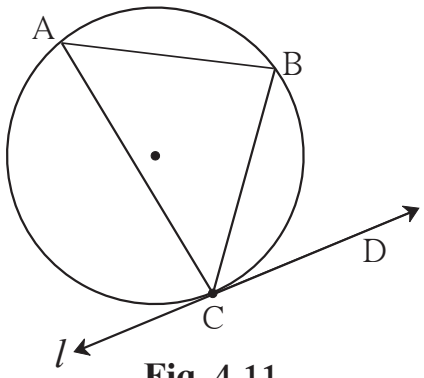


Fig. 4.11

Analysis :

As shown in the figure, let line l be the tangent to the circle at point C. Line CB is a chord and $\angle$ CAB is an inscribed angle. Now by tangent- secant angle theorem, $\angle CAB \cong \angle BCD$.

By converse of tangent- secant theorem, if we draw the line CD such that, $\angle CAB \cong \angle BCD$, then it will be the required tangent.

Steps of Construction :

(1) Draw a circle of a suitable radius. Take any point C on it.

(2) Draw chord CB and an inscribed $\angle$CAB .

(3) With the centre A and any convenient radius draw an arc intersecting the sides of $\angle$BAC in points M and N.

(4) Using the same radius and centre C, draw an arc intersecting the chord CB at point R.

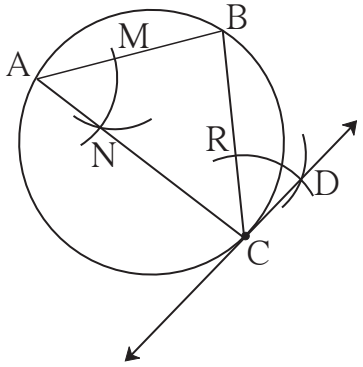


Fig. 4.12

(5) Taking the radius equal to d(MN) and centre R, draw an arc intersecting the arc drawn in the previous step. Let D be the point of intersection of these arcs. Draw line CD. Line CD is the required tangent to the circle.

Note that $\angle$ MAN and $\angle$ BCD in the above figure are congruent. If we draw seg MN and seg RD, then Δ MAN and Δ RCD are congruent by SSS test.

$\therefore \angle MAN \cong \angle BCD$

To construct tangents to a circle from a point outside the circle.

Analysis :

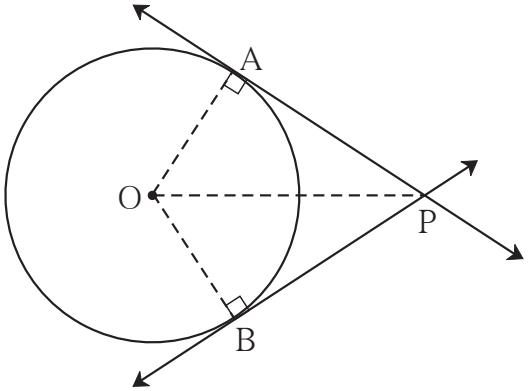


Fig. 4.13

As shown in the figure let P be a point in the exterior of the circle. Let PA and PB be the tangents to the circle with the centre O, touching the circle in points A and B respectively. So if we find points A and B on the circle, we can construct the tangents PA and PB. If OA and OB are the radii of the circle, then OA ⊥ line PA and seg OB ⊥ line PB.

Δ OAP and OBP are right angled triangles and seg OP is their common hypotenuse. If we draw a circle with diameter OP, then the points where it intersects the circle with centre O, will be the positions of points A and B respectively, because angle inscribed in a semicircle is a right angle.

Steps of Construction

(1) Construct a circle of any radius with centre O.

(2) Take any point P in the exterior of the circle.

(3) Draw segment OP. Draw perpendicular bisector of seg OP to get its midpoint M.

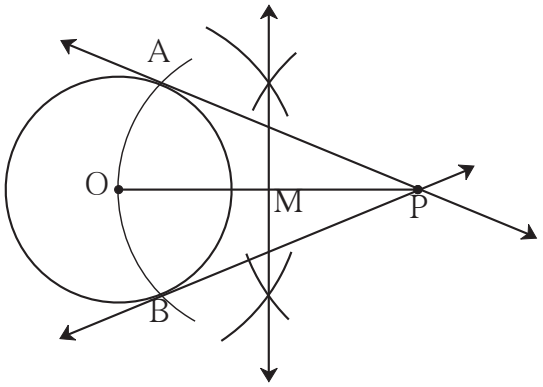


Fig. 4.14

(4) Draw a circle with radius OM and centre M

(5) Name the points of intersection of the two circles as A and B.

(6) Draw line PA and line PB.

Practice set 4.2

1. Construct a tangent to a circle with centre P and radius 3.2 cm at any point M on it.

2. Draw a circle of radius 2.7 cm. Draw a tangent to the circle at any point on it.

3. Draw a circle of radius 3.6 cm. Draw a tangent to the circle at any point on it without using the centre.

4. Draw a circle of radius 3.3 cm Draw a chord PQ of length 6.6 cm. Draw tangents to the circle at points P and Q. Write your obeservation about the tangents.

5. Draw a circle with radius 3.4 cm. Draw a chord MN of length 5.7 cm in it. construct tangents at point M and N to the circle.

6. Draw a circle with centre P and radius 3.4 cm. Take point Q at a distance 5.5 cm from the centre. Construct tangents to the circle from point Q.

7. Draw a circle with radius 4.1 cm. Construct tangents to the circle from a point at a distance 7.3 cm from the centre.

Problem set 4

1. Select the correct alternative for each of the following questions.

(1) The number of tangents that can be drawn to a circle at a point on the circle is ..... .

(A) 3 (B) 2 (C) 1 (D) 0

(2) The maximun number of tangents that can be drawn to a circle from a point out side it is ..... .

(A) 2 (B) 1 (C) one and only one (D) 0

(3) If $\Delta ABC \sim \Delta PQR$ and $\frac{AB}{PQ} = \frac{7}{5}$, then .....

(A) Δ ABC is bigger. (B) Δ PQR is bigger.

(C) Both triangles will be equal. (D) Can not be decided.

2. Draw a circle with centre O and radius 3.5 cm. Take point P at a distance 5.7 cm from the centre. Draw tangents to the circle from point P.

3. Draw any circle. Take any point A on it and construct tangent at A without using the centre of the circle.

4. Draw a circle of diameter 6.4 cm. Take a point R at a distance equal to its diameter from the centre. Draw tangents from point R.

5. Draw a circle with centre P. Draw an arc AB of 100° measure. Draw tangents to the circle at point A and point B.

6. Draw a circle of radius 3.4 cm and centre E. Take a point F on the circle. Take another point A such that E-F-A and FA = 4.1 cm. Draw tangents to the circle from point A.

7. Δ ABC $\sim$ ΔLBN. In Δ ABC, AB = 5.1cm, $\angle B = 40°$, BC=4.8 cm, $\frac{AC}{LN} = \frac{4}{7}$. Construct Δ ABC and ΔLBN.

8. Construct Δ PYQ such that, PY = 6.3 cm, YQ = 7.2 cm, PQ = 5.8 cm. If $\frac{YZ}{YQ} = \frac{6}{5}$, then construct Δ XYZ similar to Δ PYQ.

PUN6L7

5 Co-ordinate Geometry

Let's study.

- Distance formula
- Section formula
- Slope of a line

Let's recall.

We know how to find the distance between any two points on a number line.

If co-ordinates of points P,Q and R are $-1, -5$ and 4 respectively then find the length of seg PQ, seg QR.

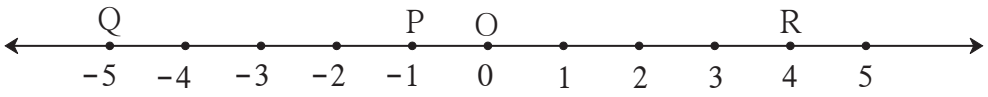


Fig. 5.1

If x_1 and x_2 are the co-ordinates of points A and B and $x_2 > x_1$ then length of seg AB = d(A,B) = $x_2 - x_1$

As shown in the figure, co-ordinates of points P,Q and R are $-1, -5$ and 4 respectively.

$\therefore$ d(P, Q) = $(-1)-(-5) = -1 + 5 = 4$

and d(Q, R) = $4 - (-5) = 4 + 5 = 9$

Using the same concept we can find the distance between two points on the same axis in XY-plane.

Let's learn.

(1) To find distance between any two points on an axis .

Two points on an axis are like two points on the number line. Note that points on the X-axis have co-ordinates such as $(2, 0)$, $(\frac{-5}{2}, 0)$, $(8, 0)$. Similarly points on the Y-axis have co-ordinates such as $(0, 1)$, $(0, \frac{17}{2})$, $(0, -3)$. Part of the X-axis which shows negative co-ordinates is OX' and part of the Y-axis which shows negative co-ordinates is OY'.

i) To find distance between two points on X-axis.

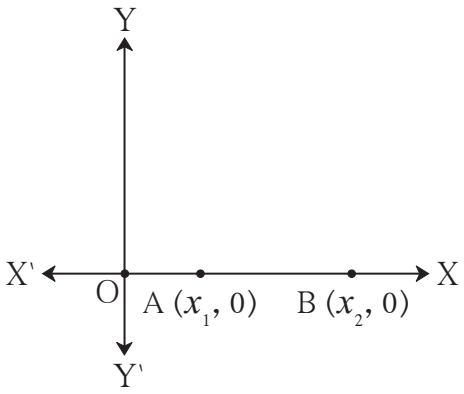


Fig. 5.2

In the above figure, points $A(x_1, 0)$ and $B(x_2, 0)$ are on X- axis such that, $x_2 > x_1$

$\therefore$ d(A, B) = $x_2 - x_1$

ii) To find distance between two points on Y-axis.

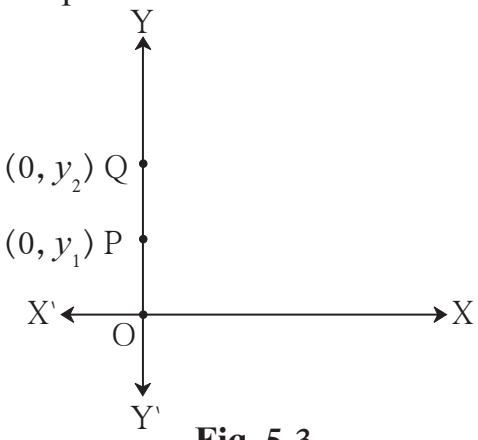


Fig. 5.3

In the above figure, points $P(0, y_1)$ and $Q(0, y_2)$ are on Y- axis such that, $y_2 > y_1$

$\therefore$ d(P,Q) = $y_2 - y_1$

(2) To find the distance between two points if the segment joining these points is parallel to any axis in the XY plane.

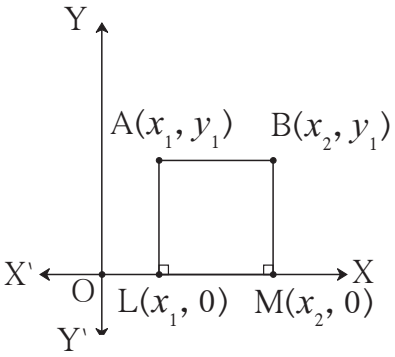


Fig. 5.4

i) In the figure, seg AB is parallel to X- axis.

$\therefore$ y co-ordinates of points A and B are equal

Draw seg AL and seg BM perpendicular to X-axis

$\therefore$ $\square$ ABML is a rectangle.

$\therefore$ AB = LM

But, LM = $x_2 - x_1$

$\therefore$ d(A,B) = $x_2 - x_1$

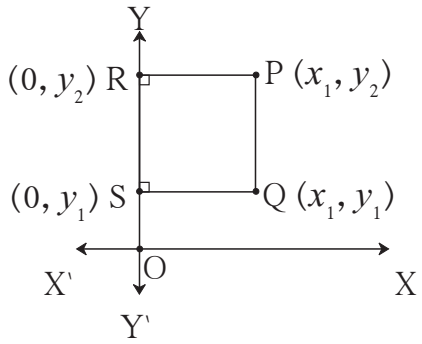


Fig. 5.5

ii) In the figure seg PQ is parallel to Y- axis.

$\therefore$ x co-ordinates of points P and Q are equal

Draw seg PR and seg QS perpendicular to Y-axis.

$\therefore$ $\square$ PQSR is a rectangle

$\therefore$ PQ = RS

But, RS = $y_2 - y_1$

$\therefore$ d(P,Q) = $y_2 - y_1$

Activity:

In the figure, seg AB || Y-axis and seg CB || X-axis. Co-ordinates of points A and C are given.

To find AC, fill in the boxes given below.

Δ ABC is a right angled triangle.

According to Pythagoras theorem,

$(AB)^2 + (BC)^2 =$ ☐

We will find co-ordinates of point B to find the lengths AB and BC,

CB || X- axis $\therefore$ y co-ordinate of B = ☐

BA || Y- axis $\therefore$ x co-ordinate of B = ☐

AB = 3 - ☐ = ☐ BC = ☐ - ☐ = 4

$\therefore$ AC^2 = ☐ + ☐ = ☐ $\therefore$ AC = $\sqrt{17}$

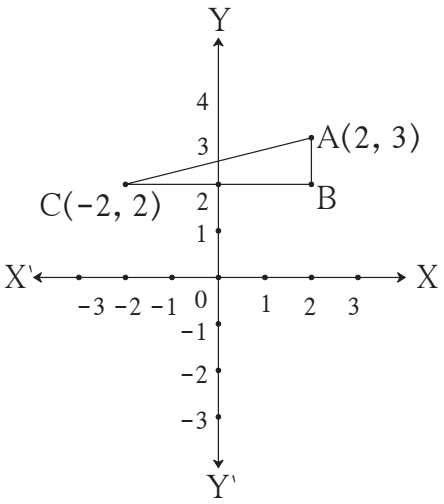


Fig. 5.6

Let's learn.

Distance formula

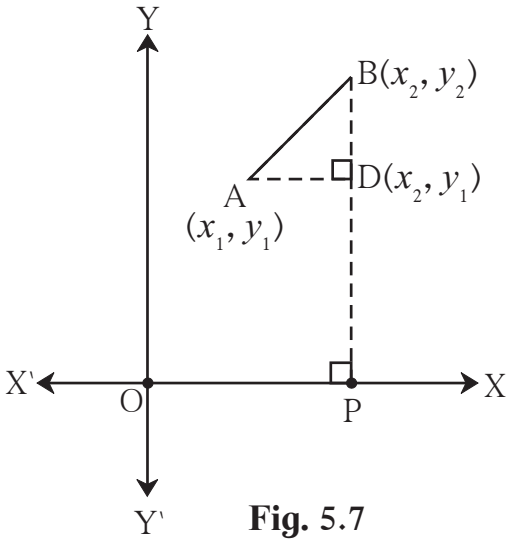


Fig. 5.7

In the figure 5.7, A(x_1, y_1) and B(x_2, y_2) are any two points in the XY plane.

From point B draw perpendicular BP on X-axis.

Similarly from point A draw perpendicular AD on seg BP.

seg BP is parallel to Y-axis.

$\therefore$ the x co-ordinate of point D is x_2.

seg AD is parallel to X-axis.

$\therefore$ the y co-ordinate of point D is y_1.

$\therefore$ AD = d(A, D) = $x_2 - x_1$; BD = d(B, D) = $y_2 - y_1$

In right angled triangle Δ ABD, $AB^2 = AD^2 + BD^2$

$= (x_2 - x_1)^2 + (y_2 - y_1)^2$

$\therefore$ AB = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

This is known as distance formula.

Note that, $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2} = \sqrt{(x_1-x_2)^2+(y_1-y_2)^2}$

In the previous activity, we found the lengths of seg AB and seg AC and then used Pythagoras theorem to find the length of seg AC.
Now we will use distance formula to find AC.

A(2, 3) and C(−2, 2) is given

Let A(x_1, y_1) and C(x_2, y_2).

$x_1 = 2, y_1 = 3, x_2 = -2, y_2 = 2$

AC $= \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

$= \sqrt{(-2-2)^2+(2-3)^2}$

$= \sqrt{(-4)^2+(-1)^2}$

$= \sqrt{16+1}$

$= \sqrt{17}$

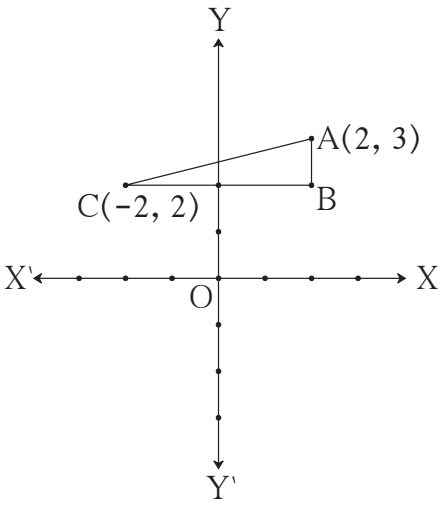


Fig. 5.8

seg AB || Y−axis and seg BC || X−axis.

∴ co-ordinates of point B are (2, 2).

∴ AB $= \sqrt{(x_2-x_1)^2+(y_2-y_1)^2} = \sqrt{(2-2)^2+(2-3)^2} = \sqrt{0+1} = 1$

BC $= \sqrt{(x_2-x_1)^2+(y_2-y_1)^2} = \sqrt{(-2-2)^2+(2-2)^2} = \sqrt{(-4)^2+0} = 4$

In the Figure 5.1, distance between points P and Q is found as (−1) − (−5) = 4. In XY- plane co-ordinates of these points are (−1, 0) and (−5, 0). Verify that, using the distance formula we get the same answer.



Remember this!

- Co-ordinates of origin are (0, 0). Hence if co-ordinates of point P are (x, y) then d(O, P) $= \sqrt{x^2+y^2}$.
- If points P(x_1, y_1), Q(x_2, y_2) lie on the XY plane then

  d(P, Q) $= \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

  that is, $PQ^2 = (x_2-x_1)^2+(y_2-y_1)^2 = (x_1-x_2)^2+(y_1-y_2)^2$

Solved Examples

Ex. (1) Find the distance between the points P(−1, 1) and Q (5,−7) .

Solution :Suppose co-ordinates of point P are (x_1, y_1) and of point Q are (x_2, y_2).

$x_1 = -1, \quad y_1 = 1, \quad x_2 = 5, \quad y_2 = -7$

According to distance formula, $d(P, Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$= \sqrt{[5-(-1)]^2 + [(-7)-1]^2}$

$= \sqrt{(6)^2 + (-8)^2}$

$= \sqrt{36+64}$

$d(P, Q) = \sqrt{100} = 10$

$\therefore$ distance between points P and Q is 10.

Ex. (2) Show that points A(−3, 2), B(1, −2) and C(9, −10) are collinear.

Solution : If the sum of any two distances out of d(A, B), d(B, C) and d(A, C) is equal to the third , then the three points A, B and C are collinear.

$\therefore$ we will find d(A, B), d(B, C) and d(A, C).

Co-ordinates of A	Co-ordinates of B	Distance formula
(−3, 2)	(1, −2)	$d(A,B) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
(x_1, y_1)	(x_2, y_2)	

$\therefore$ $d(A, B) = \sqrt{[1-(-3)]^2 + [(-2)-2]^2}$ ..... from distance formula

$= \sqrt{(1+3)^2 + (-4)^2}$

$= \sqrt{16+16}$

$= \sqrt{32} \quad = 4\sqrt{2}$ .....(I)

$d(B, C) = \sqrt{(9-1)^2 + (-10+2)^2}$

$= \sqrt{64+64} = 8\sqrt{2}$ .....(II)

and $d(A, C) = \sqrt{(9+3)^2 + (-10-2)^2}$

$= \sqrt{144+144} \quad = 12\sqrt{2}$ .....(III)

$\therefore$ from(I), (II) and (III) $4\sqrt{2} + 8\sqrt{2} = 12\sqrt{2}$

$\therefore$ d(A, B) + d(B, C) = d(A, C)

$\therefore$ Points A, B, C are collinear.

Ex. (3) Verify, whether points P(6, −6), Q(3, −7) and R(3, 3) are collinear.

Solution : PQ $= \sqrt{(6-3)^2+(-6+7)^2}$ .....by distance formula

$= \sqrt{(3)^2+(1)^2} \quad = \sqrt{10}$ ..... (I)

QR $= \sqrt{(3-3)^2+(-7-3)^2}$

$= \sqrt{(0)^2+(-10)^2} \quad = \sqrt{100}$ ..... (II)

PR $= \sqrt{(3-6)^2+(3+6)^2}$

$= \sqrt{(-3)^2+(9)^2} \quad = \sqrt{90}$ ..... (III)

From I, II and III out of $\sqrt{10}$, $\sqrt{100}$ and $\sqrt{90}$, $\sqrt{100}$ is the largest number.

Now we will verify whether $\left(\sqrt{100}\right)$ and $\left(\sqrt{10}+\sqrt{90}\right)$ are equal.

For this compare $\left(\sqrt{100}\right)^2$ and $\left(\sqrt{10}+\sqrt{90}\right)^2$.

You will see that $\left(\sqrt{10}+\sqrt{90}\right) > \left(\sqrt{100}\right)$ $\therefore$ PQ + PR $\neq$ QR

$\therefore$ points P(6, −6), Q(3, −7) and R(3, 3) are not collinear.

Ex. (4) Show that points (1, 7), (4, 2), (−1, −1) and (−4, 4) are vertices of a square.

Solution : In a quadrilateral, if all sides are of equal length and both diagonals are of equal length, then it is a square.

$\therefore$ we will find lengths of sides and diagonals by using the distance formula.

Suppose, A(1, 7), B(4, 2), C(−1, −1) and D(−4,4) are the given points.

AB $= \sqrt{(1-4)^2+(7-2)^2} \quad = \sqrt{9+25} \quad = \sqrt{34}$

BC $= \sqrt{(4+1)^2+(2+1)^2} \quad = \sqrt{25+9} \quad = \sqrt{34}$

CD $= \sqrt{(-1+4)^2+(-1-4)^2} \quad = \sqrt{9+25} \quad = \sqrt{34}$

DA $= \sqrt{(1+4)^2+(7-4)^2} \quad = \sqrt{25+9} \quad = \sqrt{34}$

AC $= \sqrt{(1+1)^2+(7+1)^2} \quad = \sqrt{4+64} \quad = \sqrt{68}$

BD $= \sqrt{(4+4)^2+(2-4)^2} \quad = \sqrt{64+4} \quad = \sqrt{68}$

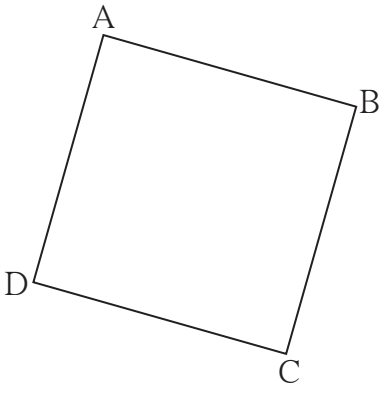


Fig. 5.9

$\therefore$ AB = BC = CD = DA and AC = BD

So, the lengths of four sides are equal and two diagonals are equal.

$\therefore$ (1,7), (4,2), (−1,−1) and (−4,4) are the vertices of a square.

Ex. (5) Find the co-ordinates of a point on Y- axis which is equidistant from M (−5,−2) and N(3,2)

Solution : Let point P(0, y) on Y- axis be equidistant from M(−5,−2) and N(3,2).

$\therefore$ PM = PN $\quad\therefore$ $PM^2 = PN^2$

$\therefore$ $[0-(-5)]^2 + [y-(-2)]^2 = (0-3)^2 + (y-2)^2$

$\therefore$ $25 + (y+2)^2 = 9 + y^2 - 4y + 4$

$\therefore$ $25 + y^2 + 4y + 4 = 13 + y^2 - 4y$

$\therefore$ $8y = -16$ $\quad\therefore$ $y = -2$

$\therefore$ the co-ordinates of the point on the Y-axis which is equidistant from M and N are M (0, −2).

Ex. (6) A(−3, −4), B(−5, 0), C(3, 0) are the vertices of Δ ABC. Find the co-ordinates of the circumcentre of Δ ABC.

Solution : Let, P(a, b) be the circumcentre of Δ ABC.

$\therefore$ point P is equidistant from A,B and C.

$\therefore$ $PA^2 = PB^2 = PC^2$ ..... (I) $\quad\therefore$ $PA^2 = PB^2$

$(a+3)^2 + (b+4)^2 = (a+5)^2 + (b-0)^2$

$\therefore a^2 + 6a + 9 + b^2 + 8b + 16 = a^2 + 10a + 25 + b^2$

$\therefore -4a + 8b = 0$

$\therefore a - 2b = 0$ ..... (II)

Similarly $PA^2 = PC^2$ ..... (I) From

$\therefore (a+3)^2 + (b+4)^2 = (a-3)^2 + (b-0)^2$

$\therefore a^2 + 6a + 9 + b^2 + 8b + 16 = a^2 - 6a + 9 + b^2$

$\therefore 12a + 8b = -16$

$\therefore 3a + 2b = -4$ ..... (III)

Solving (II) and (III) we get $a = -1$, $b = -\frac{1}{2}$

$\therefore$ co-ordinates of circumcentre are $(-1, -\frac{1}{2})$.

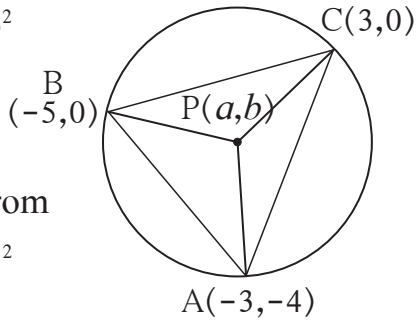


Fig. 5.10

Ex. (7) If point (x, y) is equidistant from points $(7, 1)$ and $(3, 5)$, show that $y = x-2$.

Solution : Let point P (x, y) be equidistant from points A$(7, 1)$ and B$(3, 5)$

$$\therefore \text{ AP } = \text{BP}$$

$$\therefore \text{ AP}^2 = \text{BP}^2$$

$$\therefore (x-7)^2 + (y-1)^2 = (x-3)^2 + (y-5)^2$$

$$\therefore x^2 - 14x + 49 + y^2 - 2y + 1 = x^2 - 6x + 9 + y^2 - 10y + 25$$

$$\therefore -8x + 8y = -16$$

$$\therefore x - y = 2$$

$$\therefore y = x - 2$$

Ex. (8) Find the value of y if distance between points A $(2,-2)$ and B $(-1, y)$is 5.

Solution : $\text{AB}^2 = [(-1) - 2]^2 + [y - (-2)]^2$ ..... by distance formula

$\therefore \quad 5^2 = (-3)^2 + (y + 2)^2$

$\therefore \quad 25 = 9 + (y + 2)^2$

$\therefore \quad 16 = (y + 2)^2$

$\therefore \quad y + 2 = \pm\sqrt{16}$

$\therefore \quad y + 2 = \pm 4$

$\therefore y = 4 - 2$ or $y = -4 - 2$

$\therefore y = 2$ or $y = -6$

$\therefore$ value of y is 2 or -6.

Practice set 5.1

1. Find the distance between each of the following pairs of points.

(1) A$(2, 3)$, B$(4, 1)$ (2) P$(-5, 7)$, Q$(-1, 3)$ (3) R$(0, -3)$, S$(0, \frac{5}{2})$

(4) L$(5, -8)$, M$(-7, -3)$ (5) T$(-3, 6)$, R$(9, -10)$ (6) W$(\frac{-7}{2}, 4)$, X$(11, 4)$

2. Determine whether the points are collinear.

(1) A$(1, -3)$, B$(2, -5)$, C$(-4, 7)$ (2) L$(-2, 3)$, M$(1, -3)$, N$(5, 4)$

(3) R$(0, 3)$, D$(2, 1)$, S$(3, -1)$ (4) P$(-2, 3)$, Q$(1, 2)$, R$(4, 1)$

3. Find the point on the X-axis which is equidistant from A$(-3, 4)$ and B$(1, -4)$.

4. Verify that points P$(-2, 2)$, Q$(2, 2)$ and R$(2, 7)$ are vertices of a right angled triangle.

5. Show that points P(2, −2), Q(7, 3), R(11, −1) and S (6, −6) are vertices of a parallelogram.

6. Show that points A(−4, −7), B(−1, 2), C(8, 5) and D(5, −4) are vertices of a rhombus ABCD.

7. Find x if distance between points L(x, 7) and M(1, 15) is 10.

8. Show that the points A(1, 2), B(1, 6), C($1 + 2\sqrt{3}$, 4) are vertices of an equilateral triangle.



Let's recall.

Property of intercepts made by three parallel lines :

In the figure line l || line m || line n,

line p and line q are transversals,

then $\frac{AB}{BC} = \frac{DE}{EF}$

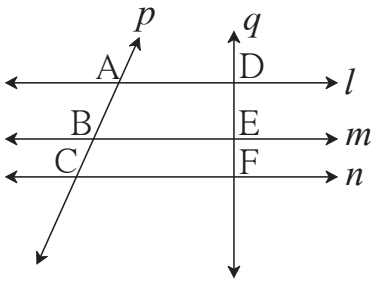


Fig. 5.11



Let's learn.

Division of a line segment

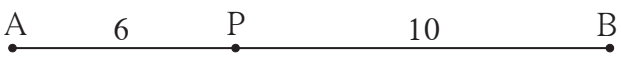


Fig. 5.12

In the figure, AP = 6 and PB = 10.

$\therefore \frac{AP}{PB} = \frac{6}{10} = \frac{3}{5}$

In other words it is said that 'point P divides the line segment AB in the ratio 3:5.

Let us see how to find the co-ordinates of a point on a segment which divides the segment in the given ratio.

Let's learn.

Section formula

In the figure 5.13, point P on the seg AB in XY plane, divides seg AB in the ratio $m : n$.

Assume $A(x_1, y_1)$ $B(x_2, y_2)$ and $P(x, y)$

Draw seg AC, seg PQ and seg BD perpendicular to X- axis.

$\therefore$ $C(x_1, 0)$; $Q(x, 0)$

and $D(x_2, 0)$.

$\therefore$ $CQ = x - x_1$

and $QD = x_2 - x$ } ..... (I)

seg AC || seg PQ || seg BD.

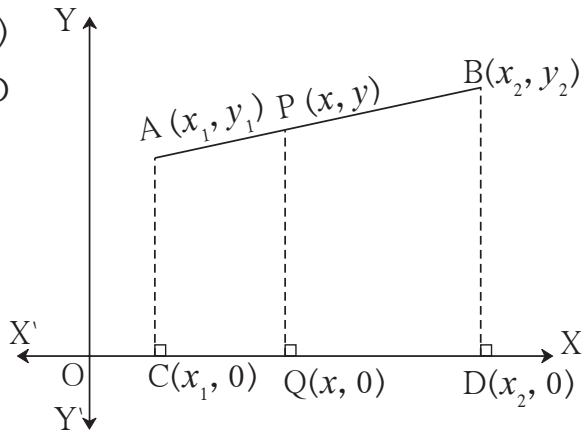


Fig. 5.13

$\therefore$ By the property of intercepts of three parallel lines, $\frac{AP}{PB} = \frac{CQ}{QD} = \frac{m}{n}$

Now $CQ = x - x_1$ and $QD = x_2 - x$ ..... from (I)

$$\therefore \frac{x - x_1}{x_2 - x} = \frac{m}{n}$$

$$\therefore n(x - x_1) = m(x_2 - x)$$

$$\therefore nx - nx_1 = mx_2 - mx$$

$$\therefore mx + nx = mx_2 + nx_1$$

$$\therefore x(m + n) = mx_2 + nx_1$$

$$\therefore x = \frac{mx_2 + nx_1}{m + n}$$

Similarly drawing perpendiculars from points A, P and B to Y- axis, we get, $y = \frac{my_2 + ny_1}{m + n}$.

$\therefore$ co-ordinates of the point, which divides the line segment joining the points $A(x_1, y_1)$ and $B(x_2, y_2)$ in the ratio $m : n$ are given by $\left(\frac{mx_2 + nx_1}{m + n}, \frac{my_2 + ny_1}{m + n}\right)$.

Co-ordinates of the midpoint of a segment

If A$(x_1, y_1)$and B(x_2, y_2) are two points and P (x, y) is the midpoint of seg AB then $m = n$.

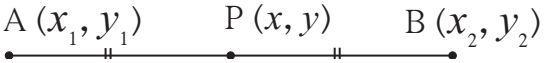


Fig. 5.14

$\therefore$ values of x and y can be written as

$$x = \frac{mx_2 + nx_1}{m+n}$$

$$= \frac{mx_2 + mx_1}{m+m} \quad \because \ m = n$$

$$= \frac{m(x_1 + x_2)}{2m}$$

$$= \frac{x_1 + x_2}{2}$$

$$y = \frac{my_2 + ny_1}{m+n}$$

$$= \frac{my_2 + my_1}{m+m} \quad \because \ m = n$$

$$= \frac{m(y_1 + y_2)}{2m}$$

$$= \frac{y_1 + y_2}{2}$$

$\therefore$ co-ordinates of midpoint P are $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

This is called as **midpoint formula**.

In the previous standard we have shown that $\frac{a+b}{2}$ is the midpoint of the segment joining two points indicating rational numbers a and b on a number line. Note that it is a special case of the above midpoint formula.

Solved Examples

Ex. (1) If A(3,5), B(7,9) and point Q divides seg AB in the ratio 2:3 then find co-ordinates of point Q.

Solution : In the given example let $(x_1, y_1) = (3, 5)$

and $(x_2, y_2) = (7, 9)$.

$m : n = 2:3$

According to section formula,

$$x = \frac{mx_2 + nx_1}{m+n} = \frac{2\times7 + 3\times3}{2+3} = \frac{23}{5} \qquad y = \frac{my_2 + ny_1}{m+n} = \frac{2\times9 + 3\times5}{2+3} = \frac{33}{5}$$

$\therefore$ Co-ordinates of Q are $\left(\frac{23}{5}, \frac{33}{5}\right)$

Ex. (2) Find the co-ordinates of point P if P is the midpoint of a line segment AB with A(−4,2) and B(6,2).

Solution : In the given example, suppose

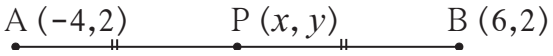


Fig. 5.15

$(-4, 2) = (x_1, y_1)$; $(6, 2) = (x_2, y_2)$ and co-ordinates of P are (x, y)

$\therefore$ according to midpoint theorem,

$$x = \frac{x_1 + x_2}{2} = \frac{-4+6}{2} = \frac{2}{2} = 1$$

$$y = \frac{y_1 + y_2}{2} = \frac{2+2}{2} = \frac{4}{2} = 2$$

$\therefore$ co-ordinates of midpoint P are (1,2) .



Let's recall.

We know that, medians of a triangle are concurrent .
The point of concurrence (centroid) divides the median in the ratio 2:1.



Let's learn.

Centroid formula

Suppose the co-ordinates of vertices of a triangle are given. Then we will find the co-ordinates of the centroid of the triangle.

In Δ ABC, $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ are the vertices. Seg AD is a median and $G(x, y)$ is the centroid.
D is the mid point of line segment BC.

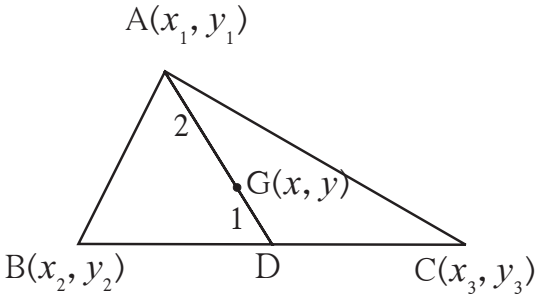


Fig. 5.16

$\therefore$ co-ordinates of point D are $x = \frac{x_2 + x_3}{2}$, $y = \frac{y_2 + y_3}{2}$ ..... midpoint theorem

Point G(x, y) is centroid of triangle Δ ABC. $\therefore$ AG : GD = 2 : 1

$\therefore$ according to section formula,

$$x = \frac{2\left(\frac{x_2 + x_3}{2}\right) + 1 \times x_1}{2+1} = \frac{x_2 + x_3 + x_1}{3} = \frac{x_1 + x_2 + x_3}{3}$$

$$y = \frac{2\left(\frac{y_2 + y_3}{2}\right) + 1 \times y_1}{2+1} = \frac{y_2 + y_3 + y_1}{3} = \frac{y_1 + y_2 + y_3}{3}$$

Thus if (x_1, y_1), (x_2, y_2) and $(x_3, y_3)$are the vertices of a triangle then the co-ordinates of the centroid are $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$.

This is called the **centroid formula.**



Remember this!

- Section formula

  The co-ordinates of a point which divides the line segment joined by two distinct points (x_1, y_1) and (x_2, y_2) in the ratio $m : n$ are $\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$.

- Midpoint formula

  The co-ordinates of midpoint of a line segment joining two distinct points (x_1, y_1) and (x_2, y_2) are $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$.

- Centroid formula

  If (x_1, y_1), (x_2, y_2) and (x_3, y_3) are the vertices of a triangle then co-ordinates of the centroid are $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$.

Solved Examples

Ex. (1) If point T divides the segment AB with A(−7,4) and B(−6,−5) in the ratio 7:2, find the co-ordinates of T.

Solution : Let the co-ordinates of T be (x, y).

$\therefore$ by the section formula,

$$x = \frac{mx_2 + nx_1}{m+n} = \frac{7\times(-6)+2\times(-7)}{7+2}$$

$$= \frac{-42-14}{9} = \frac{-56}{9}$$

$$y = \frac{my_2 + ny_1}{m+n} = \frac{7\times(-5)+2\times(4)}{7+2}$$

$$= \frac{-35+8}{9} = \frac{-27}{9} = -3$$

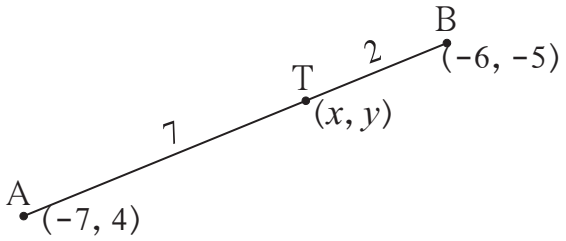


Fig. 5.17

$\therefore$ co-ordinates of point T are $\left(\frac{-56}{9}, -3\right)$.

Ex. (2) If point P(−4, 6) divides the line segment AB with A(−6, 10) and B(r, s) in the ratio 2:1, find the co-ordinates of B.

Solution : By section formula

$$-4 = \frac{2\times r + 1\times(-6)}{2+1}$$

$\therefore \quad -4 = \frac{2r-6}{3}$

$\therefore \quad -12 = 2r - 6$

$\therefore \quad 2r = -6$

$\therefore \quad r = -3$

$$6 = \frac{2\times s + 1\times 10}{2+1}$$

$\therefore \quad 6 = \frac{2s+10}{3}$

$\therefore \quad 18 = 2s + 10$

$\therefore \quad 2s = 8$

$\therefore \quad s = 4$

$\therefore$ co-ordinates of point B are (−3, 4).

Ex. (3) A(15,5), B(9,20) and A−P−B. Find the ratio in which point P(11,15) divides segment AB.

Solution : Suppose, point P(11,15) divides segment AB in the ratio $m : n$

$\therefore$ by section formula,